NEET UG High-Yield Tutorial | Biology • Class 12 • Unit IX: Biotechnology

Target Standard: NEET UG • 2–3 Questions (8–12 Marks)

Biotechnology and Its Applications

An authoritative, self-contained learning module and assessment suite structured strictly according to the NCERT core curriculum and recent NTA NEET UG examination patterns. This guide provides comprehensive conceptual pedagogy, detailed biochemical pathways, high-resolution inline vector schematics, worked analytical problem scenarios, and 20 original NEET UG practice questions with full reasoning.

1. Introduction and Critical Research Areas

Biotechnology encompasses industrial-scale bioprocessing and recombinant techniques utilized to synthesize biopharmaceuticals, biologicals, and improved agricultural commodities using genetically modified microorganisms, fungi, plants, and animals. The applications of biotechnology extend across therapeutics, molecular diagnostics, genetically modified crops in agriculture, processed food development, environmental bioremediation, waste treatment systems, and renewable bioenergy production.

Three Critical Research Areas of Biotechnology

Modern commercial biotechnology relies upon three core research thrusts:

1. **Providing the Best Catalyst:** Engineering or isolating superior biological catalysts, typically in the form of an improved microbial strain or a purified, highly active enzyme.
2. **Process Engineering:** Establishing optimal physical and chemical reaction environments (pH, temperature, substrate concentration, dissolved oxygen) through advanced biochemical engineering to maximize catalytic performance.
3. **Downstream Processing Technologies:** Formulating efficient, scalable extraction, isolation, and purification procedures to obtain the desired protein or organic metabolite with clinical-grade purity.

2. Biotechnological Applications in Agriculture

Global food demand has driven the evolution of three fundamental agricultural paradigms:

- **Agrochemical-Based Agriculture:** Driven by synthetic chemical fertilizers, herbicides, and insecticides. While the *Green Revolution* successfully tripled grain yields, this gain was predominantly achieved through improved agronomic management practices and heavy agrochemical applications. However, agrochemicals are cost-prohibitive for smallholder farmers in developing countries, conventional breeding has reached a physiological yield ceiling, and chemical runoff inflicts profound ecological damage.
- **Organic Agriculture:** Relies on natural organic manures, compost, biofertilizers, and biological pest control, avoiding synthetic agrochemicals but frequently resulting in lower or variable yields.
- **Genetically Engineered Crop-Based Agriculture:** Leverages recombinant DNA technology to generate **Genetically Modified Organisms (GMOs)**—plants, bacteria, fungi, or animals whose genetic material has been altered through purposeful molecular manipulation. GM crops provide an environmentally benign, highly specific, and sustainable agricultural alternative.

2.1 Plant Tissue Culture, Micropropagation, and Somatic Hybridization

Developed in the 1950s, **plant tissue culture** revolutionized botanical propagation when scientists demonstrated that whole plants can regenerate *in vitro* from excised plant segments under aseptic laboratory conditions.

- **Explants:** Any plant organ or excised fragment (e.g., shoot apex, leaf disc, root tip, hypocotyl, or anther) harvested and placed into a culture vessel containing a sterile nutrient formulation.

BOSONANCE

- **Totipotency:** The inherent biological capacity of an isolated plant cell or explant to divide, dedifferentiate, and regenerate into a complete, fertile adult organism.
- **Nutrient Medium Requirements:** The *in vitro* culture medium must supply:
  - A metabolizable carbon source, predominantly **sucrose**.
  - Inorganic macronutrients and micronutrients (salts).
  - Vitamins (such as thiamine, nicotinic acid, pyridoxine) and amino acids (e.g., glycine).
  - Exogenous plant growth regulators, primarily **auxins** (stimulating root morphogenesis and callus proliferation) and **cytokinins** (inducing shoot differentiation).

- **Micropropagation:** The rapid *in vitro* propagation of thousands of whole plants from explants over a brief duration under sterile conditions.
- **Somaclones:** Because micropropagation proceeds via continuous mitotic cell divisions without sexual recombination, all regenerated daughter plants are genetically identical to the parental explant and to each other. Commercially micropropagated food crops include **tomato**, **banana**, and **apple**.
- **Meristem Culture for Pathogen Eradication:** When a donor plant is systemically infected with viral pathogens, its **apical and axillary meristems remain virus-free**. This occurs because the vigorous rate of meristematic cell division outpaces viral replication, and the lack of differentiated vascular tissue (phloem) restricts systemic viral transport. By excising the meristematic dome and culturing it *in vitro*, healthy, virus-free plants are successfully produced. This technique is standard in **banana**, **sugarcane**, and **potato**.
- **Somatic Hybridization and Naked Protoplasts:** Enzymatic digestion of plant cell walls using cellulase and pectinase releases isolated, spherical **naked protoplasts** bounded solely by plasma membranes. Protoplasts derived from two different plant cultivars—each exhibiting desirable genetic attributes—can be induced to fuse, yielding a hybrid protoplast that regenerates into a **somatic hybrid**. A prominent experimental example is the **pomato**, generated by fusing tomato and potato protoplasts. Although successfully formed, the pomato lacked the full commercial agronomic traits required for widespread commercialization.

2.2 Multifaceted Agricultural Benefits of GM Plants

Agronomic Modification	Underlying Physiological Mechanism	Impact & Model Examples
Abiotic Stress Tolerance	Expression of stress-responsive proteins, osmolytes, and antioxidant defense enzymes.	Enhanced resilience against drought, extreme cold, soil salinity, and heat shock.
Pest Resistance	Endogenous expression of biological insecticidal crystal toxins without external chemicals.	Bt cotton, Bt corn; drastically reduces chemical pesticide usage.
Reduced Post-Harvest Losses	Downregulation of endogenous fruit-softening enzymes (e.g., polygalacturonase).	Extended shelf-life and enhanced shipping endurance.
Enhanced Mineral Efficiency	Engineered nutrient uptake systems preventing rapid depletion of topsoil minerals.	Sustains long-term agricultural soil fertility.
Biofortification (Nutritional Enhancement)	Engineered biosynthetic pathways producing essential vitamins in staple grains.	**Golden Rice**: Provitamin 'A' (beta-carotene) enriched rice to prevent childhood blindness.
Tailor-Made Industrial Resources	Reprogramming plant metabolism to supply alternative raw industrial feedstocks.	Customized plants producing industrial starches, biofuels, and specialized pharmaceuticals.

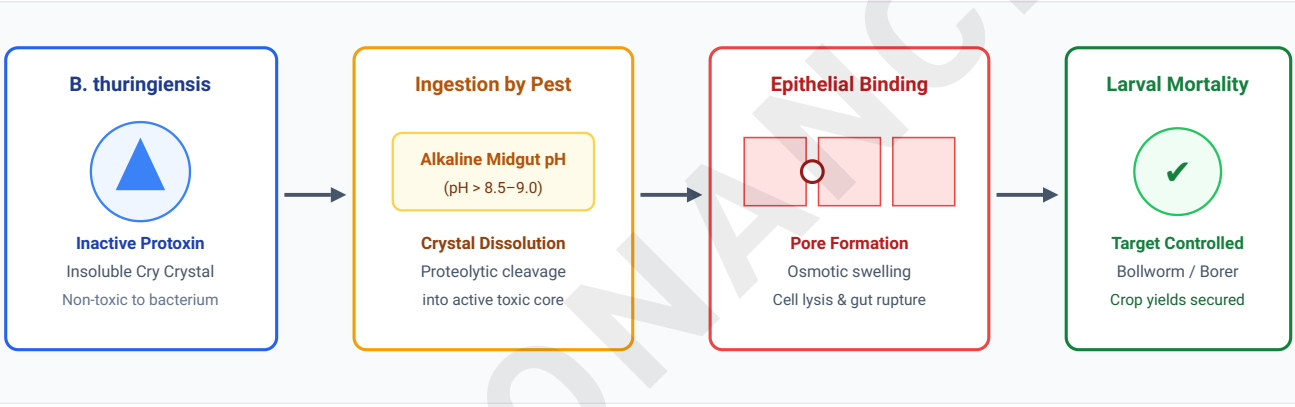
2.3 Pest-Resistant Crops: The Bt Cotton Mechanism

The ubiquitous soil bacterium *Bacillus thuringiensis* (abbreviated as **Bt**) produces specialized crystalline protein inclusions during its sporulation phase. These protein crystals contain highly potent **insecticidal proteins** (Cry proteins). Transferring these bacterial genes into crops creates endogenous resistance against destructive insect pests:

- **Target Insect Orders:**

- **Lepidopterans:** Tobacco budworm (*Heliothis virescens*), armyworm (*Spodoptera* spp.).
- **Coleopterans:** Beetles.
- **Dipterans:** True flies and mosquitoes.

- **Molecular Activation Pathway:** In the bacterium, the toxin exists as an inactive, insoluble **protoxin crystal**. This explains why the bacterium is completely unaffected. When an insect pest feeds on transgenic plant tissues, the protoxin reaches the midgut. The **alkaline pH of the insect midgut** solubilizes the crystal matrix. Midgut endoproteases then cleave the protoxin into an activated toxic core.
- **Histopathology and Insect Death:** The activated toxin binds specifically to surface receptors on **midgut epithelial cells**. Upon binding, toxin monomers oligomerize and insert into the plasma membrane, forming open **lytic pores**. These pores induce massive osmotic ion influx and water entry, leading to epithelial cell swelling, membrane lysis, gut perforation, and death of the insect larva.
- **Gene Specificity:** The insecticidal proteins are encoded by *cry* genes. Different *cry* genes confer protection against distinct insect groups:
  - `cryIAc` and `cryIIAb`: Impart specific resistance against **cotton bollworms**.
  - `cryIAb`: Imparts specific resistance against the **corn borer**.



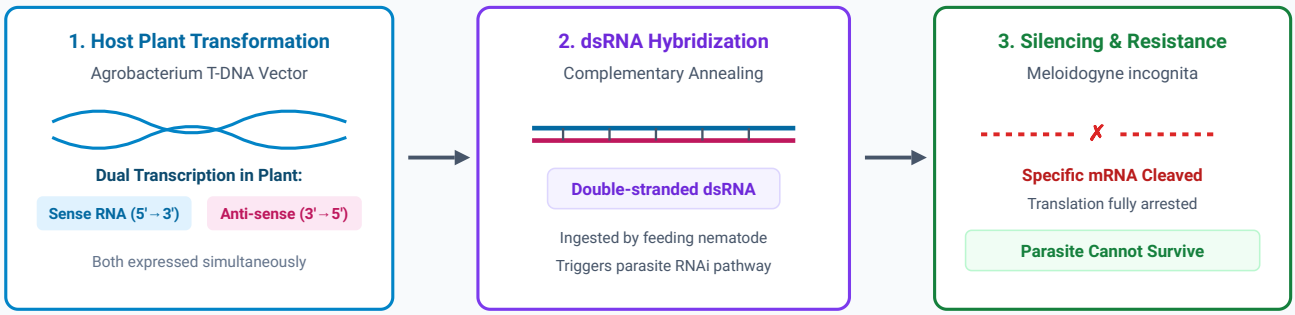
Stepwise activation and lytic pathology of insecticidal Cry protoxin in susceptible insect pests.

BOSONANCE

2.4 Pest-Resistant Plants: RNA Interference (RNAi)

The parasitic root-knot nematode *Meloidogyne incognita* attacks the root systems of tobacco plants (*Nicotiana tabacum*), inducing root galls and severely curtailing agricultural productivity. To establish resistance, researchers deployed a host-mediated gene-silencing technology termed **RNA interference (RNAi)**.

- **Evolutionary Ubiquity:** RNAi operates in **all eukaryotic organisms** as a natural cellular defense mechanism targeting foreign viral pathogens and mobile genetic elements (**transposons**) that replicate via RNA intermediates.
- **Silencing Principle:** RNAi functions by introducing a double-stranded RNA (**dsRNA**) trigger possessing sequence complementarity to a specific cellular messenger RNA (**mRNA**). The cellular RNAi machinery binds and degrades this target mRNA, blocking translation and **silencing** the gene.
- **Recombinant Engineering via Agrobacterium:** Utilizing *Agrobacterium tumefaciens* T-DNA vectors, nematode-specific cDNA sequences were introduced into the tobacco genome. The expression cassette was engineered to transcribe **both sense and anti-sense RNA strands** in the host plant cells.
- **Double-Stranded RNA Generation and Parasite Control:** Because the sense and anti-sense RNA strands were complementary, they hybridized to form double-stranded RNA (**dsRNA**). When the nematode ingested sap from the transgenic roots, the host-derived dsRNA entered nematode cells, initiating the degradation of vital nematode transcripts. Consequently, the nematode could not survive, conferring robust, durable protection to the plant.



Molecular mechanism of host-induced RNA interference silencing essential transcripts in parasitic nematodes.

3. Biotechnological Applications in Medicine

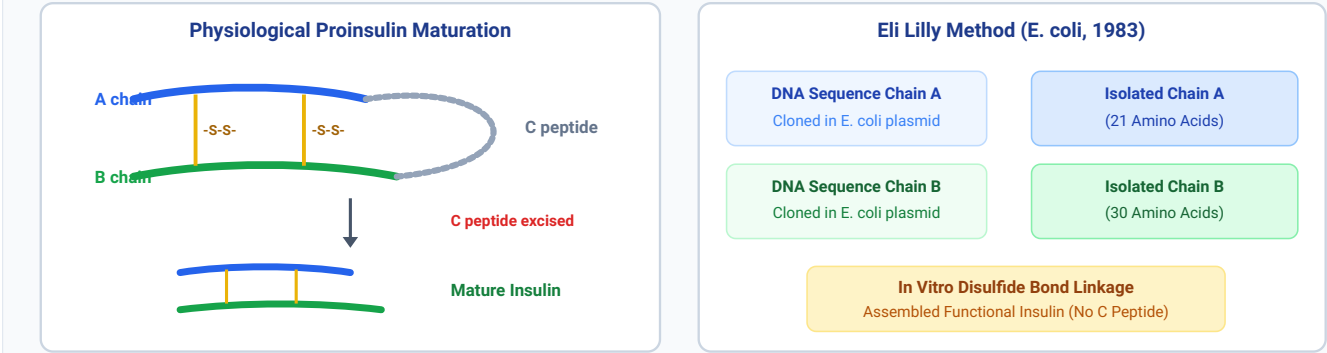
Recombinant DNA technology has enabled the mass synthesis of pure, clinically efficacious therapeutic proteins. Therapeutic proteins isolated from non-human animal tissues historically carried risks of hypersensitivity, allergic reactions, and pathogen transmission. In contrast, recombinant human therapeutics **do not elicit unwanted immunological responses** because their amino acid sequences are identical to endogenous human proteins. To date, approximately **30 recombinant therapeutics** have received global approval for human clinical use, with **12** currently marketed and distributed across India.

3.1 Genetically Engineered Human Insulin (Humulin)

Insulin is an essential peptide hormone required for glucose homeostasis in diabetes mellitus. Because insulin is a polypeptide, it **cannot be administered orally**. If swallowed, gastric hydrochloric acid and proteolytic digestive enzymes (pepsin, trypsin, and chymotrypsin) would rapidly hydrolyze its peptide bonds into constituent amino acids, abolishing its hormonal activity.

- **Historic Limitations:** Commercial insulin was previously extracted from the pancreases of slaughtered cattle and pigs. Non-human insulins triggered immune-mediated allergies, local inflammatory reactions, and anti-insulin neutralizing antibodies in sensitive diabetic patients.
- **Structure of Human Proinsulin:** In human pancreatic beta-cells, insulin is initially synthesized as an inactive precursor called **proinsulin** (pro-hormone). Proinsulin consists of three linked peptide chains:
  - **A Chain:** Consisting of 21 amino acids with an internal disulfide loop.
  - **B Chain:** Consisting of 30 amino acids.
  - **C Peptide:** An intervening connecting segment linking chains A and B.
- **Maturation:** During physiological post-translational maturation, the **C peptide is excised and eliminated**. Fully mature, biologically active human insulin consists solely of **chains A and B** connected by two interchain **disulfide bridges** ($-\mathrm{S}-\mathrm{S}-$), while a third intrachain disulfide bond stabilizes chain A. **C peptide is entirely absent in mature insulin**.
- **Recombinant Synthesis by Eli Lilly (1983):** The fundamental bioengineering hurdle in producing mature recombinant insulin in bacteria was that *Escherichia coli* lacks the mammalian endoproteases necessary to excise the C peptide and correctly fold the mature hormone. In 1983, the American pharmaceutical company **Eli Lilly** solved this challenge:
  1. Synthesized two separate DNA oligonucleotides corresponding to the human **A chain** and **B chain**.
  2. Cloned each sequence independently into plasmids of *E. coli*.
  3. Cultured both recombinant bacterial strains separately to harvest high yields of Chain A and Chain B.
  4. Extracted, purified, and chemically joined the two chains *in vitro* by establishing native disulfide bridges, successfully producing mature, fully active human insulin.

BOSONANCE



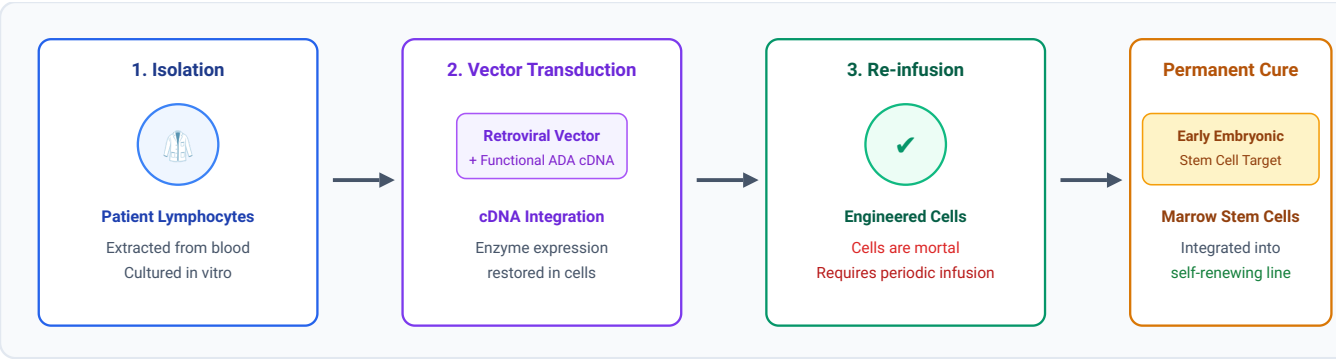
Structural maturation of proinsulin vs. the Eli Lilly recombinant manufacturing protocol.

3.2 Gene Therapy: Adenosine Deaminase (ADA) Deficiency

Gene therapy represents a collection of experimental methodologies aimed at correcting hereditary genetic mutations diagnosed in pediatric patients or human embryos. Normal, functional gene alleles are delivered into target somatic cells and tissues to compensate for and assume the function of defective or deleted genes.

- **Historic Milestone (1990):** The world's first clinical gene therapy trial was performed in **1990** on a **4-year-old girl** suffering from severe combined immunodeficiency caused by **adenosine deaminase (ADA) deficiency**.
- **Etiology and Pathophysiology:** Adenosine deaminase is an indispensable enzyme required for purine nucleotide metabolism. Deletion of the ADA gene causes intracellular accumulation of toxic purine metabolites (deoxyadenosine and dATP), which selectively arrest the proliferation of **T-lymphocytes and B-lymphocytes**, destroying both cell-mediated and humoral immunity.
- **Conventional Therapeutic Options:**
  - *Bone Marrow Transplantation:* Replaces defective hematopoietic stem cells with histocompatible donor marrow; limited by donor availability and graft-versus-host complications.
  - *Enzyme Replacement Therapy:* Regular intramuscular or intravenous injections of functional ADA enzyme.
  - *Clinical Limitation:* Neither bone marrow transplantation nor enzyme replacement therapy provides a **completely curative** outcome.
- **Ex-Vivo Retroviral Gene Therapy Protocol:**
  1. T-lymphocytes are isolated from the peripheral venous blood of the patient and cultured *in vitro* in primary culture.
  2. A functional human **ADA cDNA** construct is inserted into a non-pathogenic, replication-defective **retroviral vector**.
  3. The retroviral vector transduces the cultured lymphocytes, integrating the ADA cDNA stably into their genomes.
  4. The genetically engineered lymphocytes are returned to the patient via intravenous transfusion.
  5. **The Longevity Challenge:** Because mature peripheral lymphocytes are mortal and have a finite biological lifespan, the patient requires **periodic, lifelong infusions** of genetically modified cells.
- **Requirement for a Permanent Cure:** If the functional ADA gene isolated from marrow cells is introduced into host cells at **early embryonic stages**, it integrates into self-renewing, pluripotential stem cell lineages, providing a permanent, lifelong cure.

BOSONANCE



Clinical protocol for ADA gene therapy: Ex-vivo lymphocyte transduction vs. embryonic permanent cure.

3.3 Molecular Diagnostics

Conventional clinical diagnostic methods—such as routine serum and urine biochemical analyses—are incapable of detecting pathogens during incubation phases when microbial titers remain low. Clinical disease symptoms appear only after pathogens reach high concentrations in host tissues. Modern recombinant technologies overcome this limitation through early, ultra-sensitive detection:

- **Polymerase Chain Reaction (PCR):** Exponentially amplifies nanogram or picogram quantities of pathogen nucleic acids. PCR enables early detection long before symptoms appear:
  - Routine screening of **HIV** in suspected AIDS patients.
  - Detecting specific oncogenic point mutations, deletions, and translocations in suspected **cancer** patients.
  - Detecting inherited genetic lesions and carriers of recessive disorders.

- **Autoradiography with Radioactive Probes:** A single-stranded DNA or RNA probe tagged with a radioactive isotope is allowed to hybridize to its complementary genomic DNA in a cloned cell population, followed by exposure to photographic film:
  - Clones harboring the *normal, wild-type* gene possess exact sequence complementarity with the probe, forming stable hybrids that expose the photographic film to produce an autoradiographic band.
  - Clones harboring a **mutated gene do not appear on the photographic film** because nucleotide substitutions abolish probe complementarity, preventing hybridization and film exposure.

- **Enzyme-Linked Immunosorbent Assay (ELISA):** Founded upon the specificity of **antigen-antibody interactions**. ELISA can identify an active infection by capturing specific **pathogenic antigens** (capsular proteins, surface glycoproteins) or by detecting host **antibodies** raised against the pathogen.

4. Transgenic Animals

Transgenic animals are defined as animals whose nuclear genomes have been manipulated to carry, integrate, and express foreign (exogenous) genes. While transgenic strains of rats, rabbits, pigs, sheep, cows, and fish have been produced, **over 95 percent of all existing transgenic animals are laboratory mice**.

BOSONANCE

Domain of Application	Scientific Purpose & Biological Mechanism	Key Experimental Paradigms
1. Normal Physiology & Development	Investigating gene regulatory networks, developmental signaling pathways, and systemic endocrine roles.	Modulating **insulin-like growth factor (IGF)** expression to elucidate complex growth regulation.
2. Study of Human Disease	Serving as *in vivo* mammalian phenocopies to elucidate pathogenesis and screen experimental pharmaceuticals.	Transgenic models for **cancer**, **cystic fibrosis**, **rheumatoid arthritis**, and **Alzheimer's disease**.
3. Biological Products (Pharming)	Generating therapeutic human proteins in livestock milk or body fluids at substantially reduced production costs.	• Human **α-1-antitrypsin** for treating **emphysema**. • Therapeutic models for **phenylketonuria (PKU)** and cystic fibrosis. • **Rosie (1997)**: First transgenic cow; yielded milk containing **2.4 grams per liter** of human **α-lactalbumin**, nutritionally more balanced for human infants than cow's milk.
4. Vaccine Safety Testing	Evaluating neurovirulence and batch safety of attenuated or inactivated vaccines prior to human clinical use.	Transgenic mice developed to evaluate batch safety of **polio vaccine**, potentially replacing **monkeys**.
5. Chemical Safety & Toxicity Testing	Transgenic animals carrying genes that make them hyper-sensitive to toxic substances compared to non-transgenic controls.	Enables rapid toxicological screening of drugs and agrochemicals in significantly reduced timeframes.

5. Ethical Issues, Biopiracy, and Regulatory Frameworks

The engineering of living organisms demands strict ethical oversight. Unregulated environmental releases of genetically modified organisms could cause unpredictable ecological disruptions.

5.1 Genetic Engineering Approval Committee (GEAC)

The Indian Government constituted the **Genetic Engineering Approval Committee (GEAC)** under the Ministry of Environment, Forest and Climate Change. The statutory mandate of GEAC includes:

- Evaluating the validity, biosafety, and scientific merits of recombinant DNA research projects.
- Assessing ecological risks and approving the deliberate environmental release of GMOs and their derived products for public services.

5.2 Biopiracy and Patent Controversies

Biopiracy is defined as the unauthorized commercial exploitation of biological resources and traditional indigenous knowledge by multinational corporations or other institutions without obtaining formal authorization (Prior Informed Consent) from the source nations and without equitable compensatory payment.

- **Geopolitical Disparity:** Most industrialized nations are financially wealthy but deficient in native biodiversity and traditional ethnobotanical knowledge. Conversely, the developing and underdeveloped world is exceptionally rich in biodiversity and traditional ecological knowledge. Multinationals frequently exploit traditional knowledge to economize research time, labor, and financial capital.
- **The Basmati Rice Patent Case (1997):**
  - Rice is an ancient Asian staple. India possesses an estimated **200,000 distinct varieties of rice**, representing one of the richest germplasms globally.
  - **Basmati rice** is celebrated for its distinctive aroma, long grain, and culinary qualities. In India, **27 documented varieties of Basmati** are cultivated and recorded in historical folklore and texts.
  - In 1997, an American corporation was granted patent rights on ‘Basmati rice lines and grains’ by the US Patent and Trademark Office (USPTO). This patent authorized the company to market a purported ‘new’ variety of Basmati in the United States and international markets.

BOSONANCE

- In truth, this 'novel' variety had been generated simply by crossing an **indigenous Indian farmer's Basmati variety with a semi-dwarf cultivar**, subsequently claiming the resulting hybrid as an original invention. The patent extended to functional equivalents, threatening to restrict Indian farmers and exporters from selling authentic Basmati.

- **Exploitation of Traditional Herbal Medicines:** Similar unauthorized patent attempts have targeted traditional Indian bio-resources, notably **turmeric** (*Curcuma longa*, traditional wound-healing properties) and **neem** (*Azadirachta indica*, insecticidal and antifungal properties). Robust legal challenges by Indian authorities succeeded in revoking these illegitimate patents.
- **Legislative Protection:** To counteract biopiracy, the Indian Parliament enacted the **second amendment of the Indian Patents Bill**, incorporating specific provisions regarding patent duration, emergency licensing, and research exemptions.

6. Worked Analytical Problem Scenarios

Tutorial Analytical Scenario 1: Molecular Logic of Bt Toxin Activation

Scenario: A student hypothesizes that introducing the *cryIAc* gene into honeybees could make them resistant to bacterial pathogens. Evaluate this hypothesis based on the mechanism of Bt endotoxins.

Analytical Solution:

1. *Bacillus thuringiensis* insecticidal proteins (Cry toxins) are targeted poisons against susceptible insect herbivores (such as lepidopterans, coleopterans, and dipterans), not antibiotics against bacterial infections.
2. Cry proteins are synthesised as inactive protoxin crystals requiring an **alkaline gut environment** and specific midgut epithelial cadherin receptors to form lytic membrane pores.
3. Expressing *cryIAc* does not confer immunity against bacteria; furthermore, *cryIAc* is specifically tailored against lepidopteran larvae like cotton bollworms. The hypothesis is therefore biologically unsound.

Tutorial Analytical Scenario 2: Cellular Longevity and Gene Therapy Durability

Scenario: Explain why an ex-vivo retroviral gene therapy protocol targeting peripheral blood lymphocytes in an ADA-SCID patient requires recurring infusions, whereas delivering the construct into early embryonic cells yields a permanent cure.

Analytical Solution:

1. Peripheral T-lymphocytes are terminally differentiated somatic cells possessing a genetically finite lifespan. They are **mortal**.
2. Transfused engineered lymphocytes function for a limited time and then undergo apoptosis, causing circulating ADA enzyme levels to decline and necessitating periodic infusions.
3. A **permanent cure** is achieved when the functional ADA gene (derived from bone marrow stem cells) is integrated at **early embryonic stages**, ensuring that self-renewing, pluripotential hematopoietic stem cell lineages perpetuate and express the normal gene indefinitely.

7. NEET UG Practice Questions (Original Assessment Suite)

20 high-yield questions aligned with current NTA NEET UG blueprints (Single Correct MCQs, Assertion-Reasoning, Two-Statement Evaluations, and Matching Matrices). Full rationales and answer keys are provided.

BOSONANCE

SINGLE OPTION CORRECT MCQ

+4 / -1

Q1. Bt toxin protein does not kill *Bacillus thuringiensis* itself because:

(A) The bacterium is surrounded by an impenetrable, thick capsule

(B) The toxin exists as an inactive protoxin within the bacterium

(C) The toxin is enclosed inside specialized intracellular vacuoles

(D) The bacterial intracellular cytoplasm maintains an extremely alkaline pH

Correct Answer: (B)

Detailed Explanation: In *Bacillus thuringiensis*, the insecticidal Cry protein accumulates during sporulation as an inactive protein crystal called a **protoxin**. Because it is insoluble and enzymatically inactive at neutral/acidic intracellular pH, it cannot harm the bacterium. Conversion to the active lytic toxin requires the **alkaline pH** of the insect midgut.

BOSONANCE

MATCH THE COLUMNS

+4 / -1

Q2. Match the specific genes and biotechnological products in List-I with their corresponding target pests or applications in List-II:

List-I (Gene / Product)	List-II (Target / Application)
(a) cryIAb	(i) Cotton bollworms
(b) cryIAc	(ii) Corn borer
(c) α-1-antitrypsin	(iii) Human α-lactalbumin enriched milk
(d) Rosie the cow	(iv) Treatment of emphysema

(A) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

(B) (a)–(i), (b)–(ii), (c)–(iv), (d)–(iii)

(C) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)

(D) (a)–(i), (b)–(iv), (c)–(ii), (d)–(iii)

Correct Answer: (A)

Detailed Explanation:

- cryIAb specifically controls the **corn borer** → (a)–(ii)
- cryIAc (along with cryIIAb) controls **cotton bollworms** → (b)–(i)
- Human α-1-antitrypsin is a biological product used to treat **emphysema** → (c)–(iv)
- Rosie (first transgenic cow, 1997) produced milk enriched with human **α-lactalbumin** (2.4 g/L) → (d)–(iii)

BOSONANCE

TWO-STATEMENT EVALUATION

+4 / -1

Q3. Given below are two statements regarding RNA interference (RNAi):
Statement I: RNAi is a mechanism of cellular defense present in all eukaryotic organisms.
Statement II: In transgenic tobacco protected against *Meloidogyne incognita*, the introduced DNA produces only anti-sense RNA to block nematode translation.

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Correct Answer: (C)

Detailed Explanation: Statement I is correct: RNAi is an evolutionarily conserved cellular defense mechanism in all eukaryotes. Statement II is incorrect: *Agrobacterium*-mediated transformation was engineered such that the integrated DNA produced **both sense and anti-sense RNA** in the host cells. These complementary transcripts annealed to form double-stranded RNA (dsRNA), which triggered RNAi.

SINGLE OPTION CORRECT MCQ

+4 / -1

Q4. Which of the following structural events occurs during the post-translational maturation of human proinsulin into functional insulin?

(A) Addition of an extra C peptide to the carboxyl-terminal of chain B

(B) Removal of the intervening C peptide, leaving chains A and B linked via disulfide bridges

(C) Cleavage of the B chain into two sub-peptides joined by ester linkages

(D) Enzymatic phosphorylation of all serine residues on the A chain

Correct Answer: (B)

Detailed Explanation: Human proinsulin contains chains A, B, and C. Mature insulin consists strictly of chains A and B linked by two interchain disulfide bonds; the **C peptide is excised and eliminated** during maturation and is completely absent in mature insulin.

BOSONANCE

ASSERTION - REASON

+4 / -1

Q5. Consider the following statements:
Assertion (A): Meristem culture of the shoot apex is the preferred technique to obtain virus-free clones from a virus-infected plant.
Reason (R): The apical and axillary meristems are vascularly isolated and undergo rapid cell division that outpaces viral multiplication.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A)

(C) (A) is true, but (R) is false

(D) Both (A) and (R) are false

Correct Answer: (A)

Detailed Explanation: Even when a plant is systemically infected with a virus, its apical and axillary meristems remain free of the virus. Meristematic cells divide faster than viral replication and lack differentiated vascular channels (phloem). Excising and culturing the meristem *in vitro* successfully regenerates virus-free plants. Thus, (R) correctly explains (A).

SINGLE OPTION CORRECT MCQ

+4 / -1

Q6. In 1983, the American pharmaceutical firm Eli Lilly achieved recombinant production of human insulin by:

(A) Inserting human proinsulin cDNA into yeast cells and purifying mature insulin directly from culture broth

(B) Preparing two independent DNA sequences coding for chains A and B, expressing them separately in *E. coli*, and combining them via disulfide bonds *in vitro*

(C) Engineering transgenic dairy cattle to secrete mature human insulin in their milk

(D) Culturing pancreatic islet cells from slaughtered pigs in bioreactors

Correct Answer: (B)

Detailed Explanation: Eli Lilly prepared two distinct synthetic DNA sequences corresponding to chains A and B of human insulin and introduced them into plasmids of *E. coli*. Chains A and B were produced separately, extracted, purified, and chemically joined by forming disulfide bonds *in vitro* to reconstitute active human insulin.

BOSONANCE

TWO-STATEMENT EVALUATION

+4 / -1

Q7. Given below are two statements regarding molecular diagnostics:
Statement I: Very low concentrations of bacterial or viral pathogens can be amplified and detected by PCR before disease symptoms become clinically manifested.
Statement II: In autoradiography, a cell clone bearing a mutated gene does not appear on the photographic film when hybridized with a complementary wild-type radiolabeled probe.

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Correct Answer: (A)

Detailed Explanation: Both statements are correct. PCR amplifies scarce pathogen nucleic acid during early incubation periods (e.g., detecting HIV). In autoradiography, a radioactive probe hybridizes only with exactly complementary sequences; mutations disrupt base pairing, preventing hybridization and film exposure.

SINGLE OPTION CORRECT MCQ

+4 / -1

Q8. The enzyme Adenosine Deaminase (ADA) is critically required for the:

(A) Proper functioning and survival of immune lymphocytes

(B) Synthesis of hemoglobin in erythroblasts

(C) Hydrolysis of starch in pancreatic acinar cells

(D) Coagulation cascade in blood platelets

Correct Answer: (A)

Detailed Explanation: Adenosine deaminase plays a vital role in purine metabolism. A genetic deficiency of ADA leads to toxic accumulation of deoxyadenosine compounds, impairing the development and proliferation of T- and B-lymphocytes, culminating in severe combined immunodeficiency (SCID).

BOSONANCE

ASSERTION - REASON

+4 / -1

Q9. Consider the following statements regarding transgenic animal models:
Assertion (A): Over 95% of all existing transgenic animals are laboratory mice.
Reason (R): Mice are mammalian models with short gestation periods, high reproductive output, and well-characterized genetics that mimic human physiological pathways.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A)

(C) (A) is true, but (R) is false

(D) Both (A) and (R) are false

Correct Answer: (A)

Detailed Explanation: Over 95% of all transgenic research organisms are mice. Their small size, rapid breeding cycle, ease of genetic manipulation, and substantial physiological and genetic homology with humans make them the premier mammalian system for modeling human diseases (such as cancer, cystic fibrosis, and Alzheimer's) and testing vaccine safety.

SINGLE OPTION CORRECT MCQ

+4 / -1

Q10. How many documented varieties of Basmati rice are traditionally grown in India?

(A) 200,000

(B) 27

(C) 12

(D) 30

Correct Answer: (B)

Detailed Explanation: While India harbors an estimated 200,000 distinct varieties of rice overall, there are specifically **27 documented varieties** of Basmati grown in India, renowned worldwide for their distinct aroma, flavor, and culinary qualities.

BOSONANCE

SINGLE OPTION CORRECT MCQ

+4 / -1

Q11. The statutory Indian body responsible for assessing the validity of GM research and ensuring the biosafety of releasing GMOs for public services is:

(A) ICAR (Indian Council of Agricultural Research)

(B) CSIR (Council of Scientific and Industrial Research)

(C) GEAC (Genetic Engineering Approval Committee)

(D) IARI (Indian Agricultural Research Institute)

Correct Answer: (C)

Detailed Explanation: The Indian Government established the **GEAC** (Genetic Engineering Approval Committee) to regulate and supervise all matters concerning GM research and to evaluate the safety of introducing genetically modified organisms into the environment for public use.

MULTI-STATEMENT SELECTION MCQ

+4 / -1

Q12. Which of the following statements regarding the production of transgenic Rosie the cow is/are correct?

I. Rosie was created in 1997 as the world's first transgenic cow.

II. Her milk contained the human protein α-lactalbumin.

III. The concentration of human protein in her milk was 2.4 grams per liter.

IV. Her milk was nutritionally less balanced for human infants than normal bovine milk.

(A) I, II, and III only

(B) II and IV only

(C) I and III only

(D) I, II, III, and IV

Correct Answer: (A)

Detailed Explanation: Statements I, II, and III are correct: Rosie was produced in 1997, and her milk contained 2.4 g/L of human α-lactalbumin. Statement IV is incorrect because the presence of human α-lactalbumin made the milk **nutritionally a more balanced product** for human babies than natural cow's milk.

BOSONANCE

SINGLE OPTION CORRECT MCQ

+4 / -1

Q13. In RNA interference, the source of the complementary double-stranded RNA (dsRNA) trigger can be:

(A) Infection by viruses possessing RNA genomes

(B) Mobile genetic elements (transposons) that replicate via an RNA intermediate

(C) Agrobacterium-mediated introduction of genes producing both sense and anti-sense RNA

(D) All of the above

Correct Answer: (D)

Detailed Explanation: In nature, cellular RNAi can be triggered by infections from RNA viruses or by transposons that mobilize through RNA intermediates. In recombinant agricultural applications (such as nematode-resistant tobacco), dsRNA is artificially induced by introducing constructs via *Agrobacterium* that simultaneously generate both sense and anti-sense transcripts.

ASSERTION - REASON

+4 / -1

Q14. Consider the following statements:
Assertion (A): Therapeutic proteins produced via recombinant DNA technology are preferred over those extracted from slaughtered animal tissues.
Reason (R): Recombinant therapeutics are structurally identical to human proteins and do not trigger unwanted immunological or allergic responses.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A)

(C) (A) is true, but (R) is false

(D) Both (A) and (R) are false

Correct Answer: (A)

Detailed Explanation: Therapeutic proteins extracted from non-human animal sources (e.g., porcine or bovine insulin) often differ slightly in primary sequence, provoking allergic and immune responses. Recombinant DNA technology produces proteins identical to native human molecules, eliminating unwanted immune sensitization and the risk of pathogen transmission.

BOSONANCE

SINGLE OPTION CORRECT MCQ

+4 / -1

Q15. The term 'Biopiracy' specifically denotes:

(A) The physical poaching and illegal trade of endangered wildlife species

(B) The unauthorized commercial exploitation of bio-resources and traditional indigenous knowledge without proper authorization and compensatory sharing

(C) The unauthorized duplication and electronic distribution of biotechnology research papers

(D) Accidental contamination of wild cultivars by pollen from genetically modified crop fields

Correct Answer: (B)

Detailed Explanation: Biopiracy refers strictly to the unauthorized use of bio-resources and associated traditional knowledge by multinational organizations and institutions without obtaining formal authorization from the sovereign nation and without equitable compensatory payment to indigenous communities.

SINGLE OPTION CORRECT MCQ

+4 / -1

Q16. Somatic hybridization involves which of the following sequential procedures?

(A) Meiotic recombination between two distinct pollen grains in vitro

(B) Enzymatic digestion of plant cell walls using cellulase and pectinase, isolation of naked protoplasts, fusion of protoplasts, and regeneration into a hybrid plant

(C) Direct microinjection of isolated plasmid DNA into the nucleus of a root hair cell

(D) Chemical fusion of gametic nuclei followed by culture on hormone-free agar

Correct Answer: (B)

Detailed Explanation: Somatic hybridization involves digesting the cellulose-pectin cell walls of plant cells using cellulase and pectinase enzymes to isolate naked protoplasts bounded only by plasma membranes. Protoplasts from two distinct varieties are then fused chemically (e.g., using PEG) or electrostatically to form hybrid protoplasts, which are subsequently regenerated into whole somatic hybrid plants (e.g., pomato).

BOSONANCE

SINGLE OPTION CORRECT MCQ

+4 / -1

Q17. Which of the following components must be provided in the nutrient medium for successful in vitro plant tissue culture from an explant?

(A) Carbon source such as sucrose, inorganic salts, vitamins, amino acids, and growth regulators (auxins and cytokinins)

(B) Only distilled water, glucose, and nitrogen gas

(C) Soil extract, cellulase enzymes, and lipid emulsions

(D) Carbon dioxide gas, pectinase enzymes, and animal serum

Correct Answer: (A)

Detailed Explanation: For the successful in vitro regeneration of explants via tissue culture, the defined nutrient medium must supply a carbon source (such as sucrose), inorganic macro- and micro-salts, essential vitamins, amino acids, and key plant growth regulators (auxins to stimulate root morphogenesis and cytokinins to stimulate shoot morphogenesis).

SINGLE OPTION CORRECT MCQ

+4 / -1

Q18. Golden rice is a genetically engineered crop developed primarily to alleviate:

(A) Iron-deficiency anemia in rice-consuming populations

(B) Vitamin 'A' deficiency, by expressing provitamin 'A' (beta-carotene) in the rice endosperm

(C) Susceptibility to fungal blast disease

(D) Sensitivity to prolonged soil waterlogging and submergence

Correct Answer: (B)

Detailed Explanation: Golden rice is a genetically biofortified variety enriched with Vitamin 'A' (beta-carotene, a provitamin A carotenoid). It was specifically engineered to combat childhood blindness and severe xerophthalmia prevalent in developing nations where polished white rice is the dietary staple.

BOSONANCE

TWO-STATEMENT EVALUATION

+4 / -1

Q19. Given below are two statements regarding transgenic animal testing:
Statement I: Transgenic mice are being developed and validated to test the batch safety of polio vaccines before human administration.
Statement II: If successful, transgenic mice could replace the traditional use of monkeys for testing polio vaccine batch safety.

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Correct Answer: (A)

Detailed Explanation: Both statements are correct. Transgenic mice carrying the human poliovirus receptor gene are employed to assess the safety, neurovirulence, and batch consistency of polio vaccines. Their reliability offers an ethical and scientific alternative that replaces the use of rhesus monkeys.

BOSONANCE

MATCH THE COLUMNS

+4 / -1

Q20. Match the insect orders in List-I with their representative pests targeted by specific Bt toxins in List-II:

List-I (Insect Order)	List-II (Targeted Insect Pest)
(a) Lepidopterans	(i) Beetles
(b) Coleopterans	(ii) Flies and mosquitoes
(c) Dipterans	(iii) Tobacco budworm and armyworm

(A) (a)–(iii), (b)–(i), (c)–(ii)

(B) (a)–(i), (b)–(ii), (c)–(iii)

(C) (a)–(ii), (b)–(iii), (c)–(i)

(D) (a)–(iii), (b)–(ii), (c)–(i)

Correct Answer: (A)

Detailed Explanation:

- Lepidopterans include the **tobacco budworm** and **armyworm** → (a)–(iii)
- Coleopterans include **beetles** → (b)–(i)
- Dipterans include true **flies and mosquitoes** → (c)–(ii)

8. Common Pitfalls & High-Yield Revision Checklist

Frequent NEET Traps & Examination Distinctions

- **cry Gene vs. Cry Protein:** The gene designation is italicized and begins with a lower-case letter (e.g., `cryIAc`, `cryIAb`), whereas the expressed toxin protein is non-italicized and capitalized (**Cry protein**).
- **Bollworm vs. Corn Borer Specificity:** Cotton bollworms are controlled by `cryIAc` and `cryIIAb`. In contrast, the corn borer is controlled by `cryIAb`. Do not confuse `cryIAb` with `cryIAc`!
- **C Peptide in Insulin:** C peptide is present in the precursor molecule **proinsulin**, but is completely **absent in mature insulin**. In Eli Lilly's recombinant manufacturing process, chains A and B were synthesized independently in *E. coli* and assembled *in vitro*; proinsulin was never formed.
- **RNAi Strand Origin:** In transgenic host plants, the introduced expression vector transcribes **both sense and anti-sense RNA**, which anneal to produce double-stranded RNA (dsRNA). It is not single-stranded antisense RNA alone.
- **Longevity of Gene Therapy:** Transfusion of transduced adult peripheral lymphocytes provides only a temporary clinical treatment because mature lymphocytes are mortal. A **permanent cure** is achieved only when the functional gene is introduced into cells at **early embryonic stages**.

BOSONANCE

Bt Toxin Target Orders

- **Lepidopterans:** Tobacco budworm, armyworm
- **Coleopterans:** Beetles
- **Dipterans:** Flies, mosquitoes

Insulin Structure & Chains

- **Chain A:** 21 amino acids
- **Chain B:** 30 amino acids
- **Disulfide bridges:** 2 interchain, 1 intrachain
- **Oral delivery:** Ineffective (hydrolyzed by digestive proteases)

Approved Therapeutics & GMOs

- **Recombinant drugs:** 30 globally approved, 12 in India
- **Golden Rice:** Provitamin 'A' (beta-carotene) enriched
- **Rosie cow (1997):** 2.4 g/L human α-lactalbumin milk

ADA Gene Therapy Milestones

- **1990:** 4-year-old girl with ADA deficiency
- **Vector:** Disarmed retrovirus carrying ADA cDNA
- **Target:** Autologous cultured T-lymphocytes
- **Permanent cure:** Early embryonic bone marrow delivery

Molecular Diagnostics

- **PCR:** Nucleic acid amplification for asymptomatic titers
- **ELISA:** Antigen-antibody binding interactions
- **Autoradiography:** Mutated clones fail to expose photographic film

Bioethics & Patent Laws

- **GEAC:** Regulates GMO research and biosafety in India
- **1997:** US patent on Indian Basmati crossed with semi-dwarf
- **Indian Patents Bill:** Second amendment

9. Concise Final Recap

Biotechnology interfaces cellular genetics with large-scale industrial bioprocessing. In **agriculture**, cellular **totipotency** enables rapid micropropagation of uniform somaclones, meristem isolation yields virus-free germplasms, and recombinant transgenesis enhances abiotic stress tolerance, optimizes soil mineral usage, biofortifies staple foods (Golden Rice), and confers endogenous pest resistance through *cry* genes (Bt cotton) and RNAi (silencing *Meloidogyne incognita*). In **medicine**, recombinant engineering yields pure, non-immunogenic human proteins (such as Humulin synthesized by Eli Lilly in 1983) and facilitates corrective somatic interventions like ex-vivo retroviral gene therapy for ADA deficiency. **Molecular diagnostics** (PCR, ELISA, probe autoradiography) permit ultra-sensitive detection at asymptomatic stages. **Transgenic animals** (>95% mice) serve as physiological and disease models, living bioreactors for biological products (human α-1-antitrypsin, α-lactalbumin in Rosie), and substrates for chemical toxicity and vaccine safety testing. Finally, statutory oversight by national agencies such as the **GEAC** and proactive amendments to patent laws guard against environmental hazards and unauthorized biopiracy of sovereign bio-resources.

BOSONANCE